

Multimodal Fake News and Deepfake Detection Using Hybrid AI Models: A Modular Framework Integrating Text, Audio, Image, Video, and Lip-Sync Analysis

Aneela Pervez

Department of Computer Science, GIFT University, Gujranwala, Pakistan

Email: 241440001@gift.edu.pk

Abstract—The proliferation of generative AI has produced increasingly realistic synthetic media across text, image, audio, and video modalities, posing systemic threats to information integrity, journalism, biometric authentication, and public trust. Most existing deepfake detectors operate on a single modality and exhibit poor cross-domain generalization, while end-to-end multimodal systems often sacrifice interpretability and real-time deployability. In this paper, we present a modular multimodal fake-news and deepfake detection framework that integrates five complementary analysis streams—textual claim verification, audio spoofing detection, image forgery analysis, video manipulation detection, and audio-visual lip-sync consistency—using a hybrid CNN-Transformer-LLM architecture. The text module combines a DeBERTa cross-encoder with retrieval-augmented Llama-3.3 verification through the SERP API; the audio module fuses Wav2Vec2, AST, CLAP, and Whisper through fuzzy logic; the image module uses a ResNet-18 + 2-layer Transformer encoder; the video module employs ViT and OpenCLIP with multi-frame voting and Claude 3.5 Sonnet reasoning; and the lip-sync module uses a Lip-Sync Embedding Network (LSE-Net) for cross-modal phoneme-viseme alignment. The system is trained and evaluated on FaceForensics++, FakeAVCeleb, and a curated audio dataset combining ASVspoof samples with custom recordings. The full pipeline is deployed via Gradio on Hugging Face Spaces, providing real-time inference and explainable per-modality outputs. Experimental evaluation yields per-modality accuracy of 99% (text), 92% (image), 91% (audio), 91% (video), and 89% (lip-sync). Unlike prior end-to-end multimodal systems, our modular design enables independent module updates and produces interpretable per-stream verdicts, which we argue are essential for real-world deployment by journalists, fact-checkers, and forensic analysts. We discuss the system's limitations regarding compression artifacts, adversarial robustness, and dataset bias, and outline future work toward unified late-fusion evaluation and in-the-wild benchmarking.

Index Terms—Deepfake detection, multimodal AI, fake news detection, multimedia forensics, hybrid deep learning, transformers, large language models, trustworthy AI.

I. INTRODUCTION

Recent advances in generative artificial intelligence—particularly Generative Adversarial Networks (GANs) [1], denoising diffusion probabilistic models [2], and large multimodal foundation models—have made it possible to synthesize hyper-realistic fake images, audio, and video at unprecedented scale and quality. These technologies, collectively referred to as deepfakes, have been weaponized for cryptocurrency scams, election interference, identity fraud, and corporate espionage. In one widely reported 2024 incident, a multinational engineering firm lost approximately USD 25 million after a deepfake video conference impersonated a chief financial officer. Such incidents underscore that deepfake detection is no longer a purely academic concern; it is a deployment-critical security problem [3].

Despite substantial progress, three persistent gaps limit the practical utility of current deepfake detectors. First, the majority of published methods focus on a single modality—typically face-swap video [4] or synthetic audio [5]—and therefore fail to capture cross-modal inconsistencies that human verifiers naturally exploit, such as audio that does not match lip movement or text claims that contradict visual evidence [6], [7]. Second, monolithic end-to-end multimodal architectures, while theoretically expressive, are difficult to interpret, slow to retrain when individual modalities improve, and impractical to deploy on resource-constrained infrastructure. Third, real-world misinformation increasingly combines authentic media with false textual claims; pure pixel-level forgery detection misses this entire class of attack [8], [9].

To address these gaps, this paper proposes a modular multimodal detection framework that combines five independent analysis streams. The contributions of this work are summarized as follows:

- 1) We design a five-modality hybrid architecture combining CNN, Transformer, and LLM components, where each module addresses a distinct attack surface (textual disinformation, audio spoofing, image forgery, video manipulation, and lip-sync inconsistency).
- 2) We curate and combine three benchmark datasets (FaceForensics++ [4], FakeAVCeleb [10], and a custom audio set) and introduce a real-time text-verification pipeline integrating SERP-based evidence retrieval with LLM-based stance analysis.
- 3) We provide a publicly accessible deployment of the system on Hugging Face Spaces with explainable per-modality outputs, enabling reproducibility and downstream evaluation by the community.
- 4) We report per-modality test performance across all five streams and discuss the system's behavior, limitations, and threats to validity in detail.

Figure 1 shows the unified Gradio interface of the deployed system, exposing each modality through an independent analysis tab. The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed methodology. Section IV describes the datasets and preprocessing. Section V reports experimental results. Section VI discusses findings, Section VII outlines limitations, and Section VIII concludes.



Fig. 1. Unified Gradio interface of the proposed system, exposing per-modality analysis tabs for Text, Audio, Video, Multi-Modal Lip-Sync, and Image analysis.

II. RELATED WORK

A. Survey-Level Multimodal Studies

Lv et al. [11] provide a comprehensive survey of deep-learning techniques for multimodal fake-news detection, identifying data scarcity and limited interpretability as the principal open challenges. Liu et al. [12] survey the evolution from single-modal to multimodal facial deepfake detection, highlighting the same gap. Qureshi et al. [13] review digital forensic methods for multimodal deepfake identification on social media.

B. Unimodal Deepfake Detection

Early deepfake detection focused on detecting GAN fingerprints in face-swapped images, using XceptionNet and similar CNN backbones trained on FaceForensics++ [4]. Subsequent work introduced frequency-domain analysis [14], biological signal cues, and adversarial training. Audio-only spoofing detection has matured around the ASVspoof benchmarks [5], with state-of-the-art systems combining Wav2Vec2 [15] embeddings with classification heads.

C. Audio-Visual and Multimodal Detection

Audio-visual systems detect mismatches between lip movement and acoustic phonemes [16], [17]. Khalid et al. [10] introduced FakeAVCeleb, the standard audio-video deepfake benchmark. Alsaeedi et al. [18] propose emotion-aware multimodal fusion. Zhang et al. [19] introduce ERF-BA-TFD+, a Transformer-based audio-visual fusion architecture. Li et al. [20] focus on multimodal short-video fake-news detection, while Al-alshaqi et al. [21] propose a BERT-based text-image multimodal framework.

D. Benchmarks and Foundation-Model Detectors

Zhu et al. [22] release MFND, a large-scale multimodal benchmark. Chandra et al. [3] introduce Deepfake-Eval-2024, an in-the-wild benchmark showing that state-of-the-art open-source detectors drop by 45–50 AUC percentage points compared to academic benchmarks. Jia et al. [9] investigate whether ChatGPT-style multimodal LLMs can themselves serve as deepfake detectors.

E. Explainable Deepfake Forensics and the Detection Arms Race

A growing literature emphasizes that detection accuracy alone is insufficient: forensic analysts require interpretable evidence [23]. Verdoliva [8] argues that data-driven detectors must be combined with traditional forensic features. Croitoru et al. [24] characterize the ongoing tug-of-war between deepfake generation and detection, advocating for proactive research collaboration and standardized benchmarks. Our work follows this direction by producing per-modality decisions and LLM-generated reasoning.

F. Position of This Work

Compared to prior work, the present framework differs in three ways: (i) it is explicitly modular, allowing each module to be swapped without retraining the others; (ii) it integrates a real-time LLM-based textual verification stream alongside multimedia forensics, addressing the under-studied combination of fake-news and deepfake detection [21]; and (iii) it is deployed publicly with explainable outputs.

III. METHODOLOGY

The proposed system is organized as five parallel detection modules. Each module produces an independent verdict together with a confidence score and a natural-language reasoning trace, allowing analysts to inspect the basis of every decision. The remainder of this section describes the architecture and processing pipeline of each module.

A. Text Verification Module

The text module accepts a textual claim and produces a credibility score in $[0, 1]$. Figure 2 presents its architecture. The pipeline first retrieves contemporaneous web evidence via the Google SERP API [25], then computes cosine similarity between the claim C and each evidence snippet E_i using a DeBERTa-v3 cross-encoder [26]:

$$\text{sim}(C, E_i) = (F_C \cdot F_{\{E_i\}}) / (||F_C|| \cdot ||F_{\{E_i\}}||), \quad i = 1, \dots, n$$

Per-evidence similarities are then aggregated into a weighted evidence score:

$$S(C) = \sum_{i=1}^n a_i \cdot \text{sim}(C, E_i), \quad \sum_{i=1}^n a_i = 1.0, \quad a_i \geq 0$$

where weights a_i emphasize evidence from authoritative news sources. The strongest evidence is forwarded to a Groq Llama-3.3 reasoning module [27], which determines whether the evidence supports or refutes the claim and produces a natural-language verdict with citations to the underlying sources. Figure 3 shows a representative output on the input "Iran war with America has been stopped," yielding a FAKE/DEBUNKED verdict at 99% confidence.

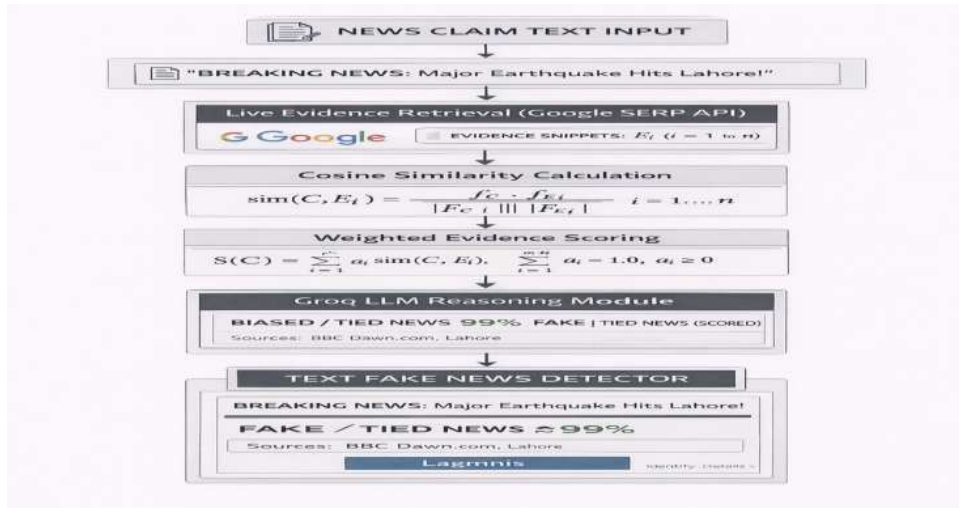


Fig. 2. Architecture of the text claim-verification module: SERP-based evidence retrieval, DeBERTa cross-encoder cosine similarity, weighted evidence scoring, and Groq Llama-3.3 reasoning.

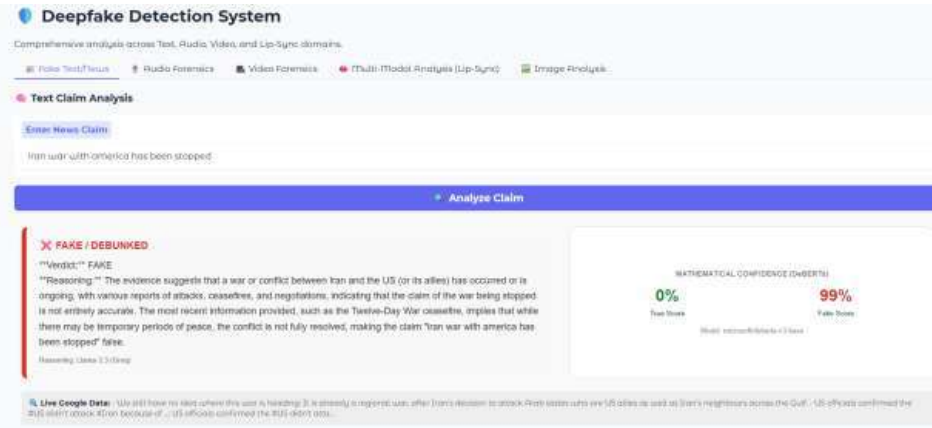


Fig. 3. Example output of the text module on the claim "Iran war with America has been stopped": FAKE/DEBUNKED verdict at 99% confidence, supported by retrieved evidence and an LLM-generated explanation.

B. Audio Detection Module

The audio module operates on raw waveforms resampled to 16 kHz (for speech models) and 48 kHz (for environmental analysis). Figure 4 shows the pipeline. Wav2Vec2 [15] produces a per-segment voice DNA fingerprint, and identity-consistency metrics (cosine similarity and Euclidean distance) verify speaker continuity. Four parallel expert engines then operate simultaneously: Whisper [28] transcribes speech and identifies linguistic anomalies; the Audio Spectrogram Transformer (AST) [29] examines microscopic spectral textures characteristic of vocoder-generated speech; CLAP [30] performs zero-shot environment matching to detect inconsistency between voice and background; and onset-strength analysis searches for splices. Expert outputs are combined through fuzzy membership calculation across three categories—Authentic, AI-Synthesized, and Manipulated/Spliced—and the final forensic verdict is generated by Llama-3.3. Figure 5 shows a representative output classifying an audio clip as AI_CLONE_DETECTED with texture score 100% and fuzzy score 1.00.

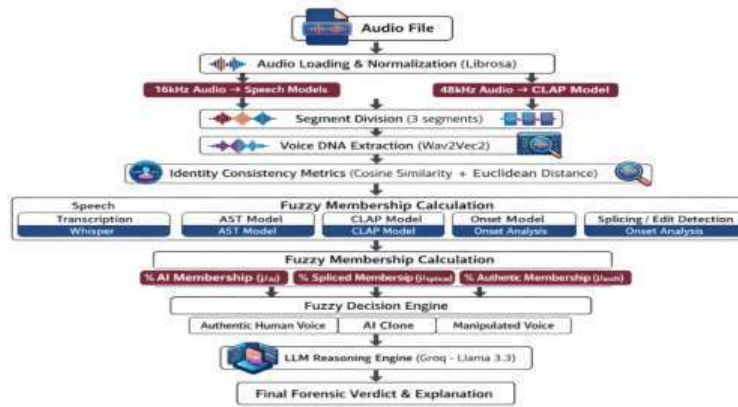


Fig. 4. Architecture of the audio deepfake detection module combining voice DNA extraction (Wav2Vec2), AI-texture analysis (AST), environment matching (CLAP), splice detection (onset analysis), and Whisper transcription, fused via fuzzy logic and finalized by Llama-3.3.

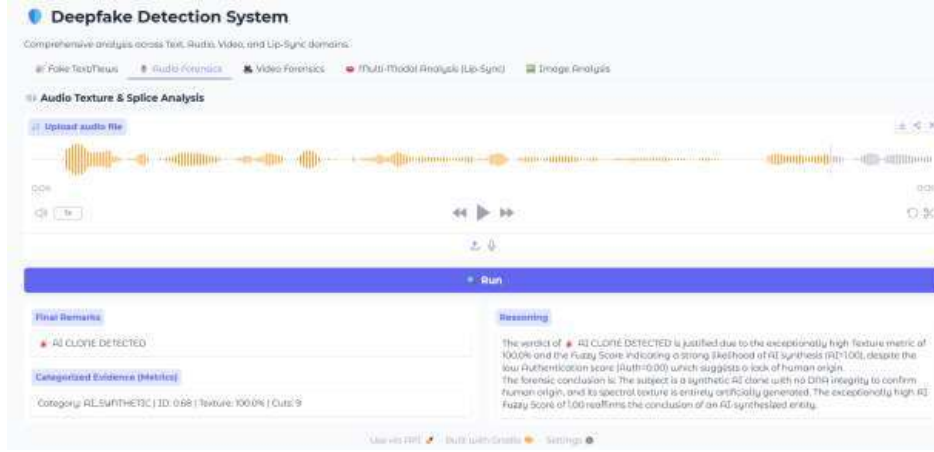


Fig. 5. Example output of the audio module: an uploaded audio clip is classified as *AI_CLONE_DETECTED* with texture 100% and fuzzy score 1.00, accompanied by a detailed forensic justification.

C. Image Detection Module

The image module accepts a single RGB frame, resized to 224×224 and normalized using standard ImageNet statistics, and produces a forgery score. Figure 6 shows the architecture. We use a hybrid backbone combining a ResNet-18 [31] CNN extractor (last classification layers removed, output [B, 512, 7, 7]) with a 2-layer Transformer encoder [32] ($d_{\text{model}}=512$, $n_{\text{head}}=8$, $d_{\text{ff}}=1024$, $\text{dropout}=0.3$). The 7×7 spatial grid is flattened into a 49-token sequence and processed with self-attention to capture global structural inconsistencies that local CNN features miss. The encoder output is mean-pooled across the sequence and passed through a classifier block (Linear $512 \rightarrow 128$, BatchNorm, ReLU, Dropout 0.4, Linear $128 \rightarrow 2$) followed by Softmax. The module is trained with class-balanced binary cross-entropy on FaceForensics++ [4]. Figure 7 shows a representative output.

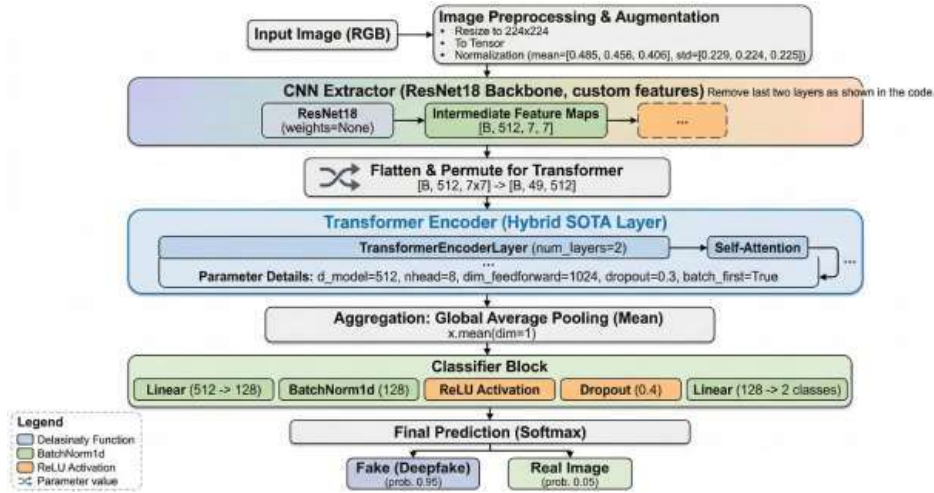


Fig. 6. Architectural workflow of the proposed hybrid CNN-Transformer network for deepfake image detection. ResNet-18 extracts local features that are flattened, processed by a 2-layer Transformer encoder with self-attention, mean-pooled, and classified via a fully-connected block with Softmax output.



Fig. 7. Example output of the image module classifying an uploaded portrait as Real Image with 100% confidence.

D. Video Detection Module

The video module processes 10 frames uniformly sampled from each input video. Figure 8 shows the pipeline. RetinaFace [33] detects and crops the dominant face in each frame, with image thumbnailing for speed. Two expert modules then operate in parallel: (i) a Hugging Face ViT model (dima806/deepfake_vs_real) votes Fake or Real on each of the 10 face crops; (ii) OpenCLIP ViT-B/32 [34] computes correlation between the frames and 15 Real-prompts plus 15 Fake-prompts, producing semantic similarity scores sim_real and sim_fake . A majority voting threshold of 6/10 Fake votes triggers the FORGED verdict. Final reasoning is produced by Claude 3.5 Sonnet, which receives the verdict, vote counts, OpenCLIP scores, and best face crop, and produces a Detailed Forensic Reasoning report. Figure 9 shows a representative output where 10/10 frames vote Fake.

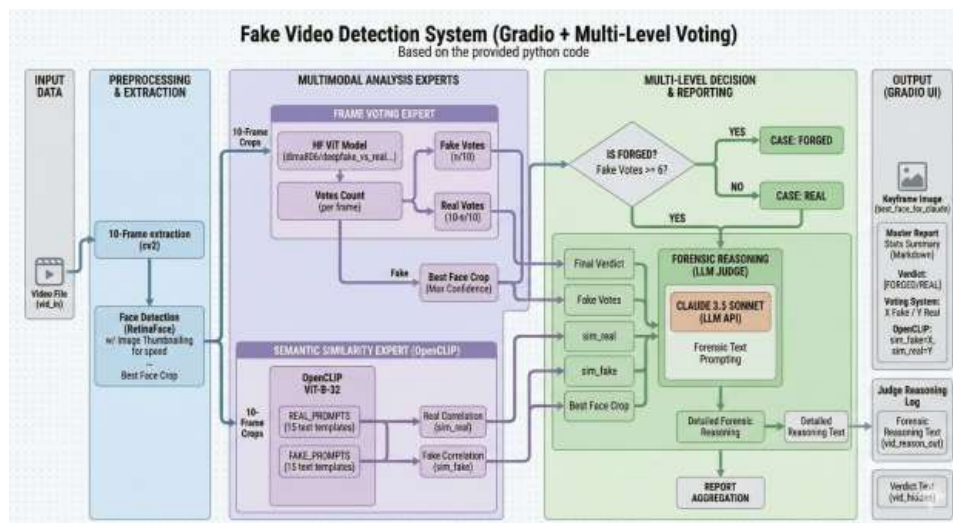


Fig. 8. Architecture of the video deepfake detection system: 10-frame extraction, RetinaFace face detection, parallel ViT-based frame voting and OpenCLIP semantic similarity, multi-level decision aggregation, and Claude 3.5 Sonnet forensic reasoning.



Fig. 9. Example output of the video module: the uploaded video is classified as FORGED with 10/10 frames voted Fake by the ViT expert and corresponding OpenCLIP semantic similarity scores.

E. Lip-Sync Detection Module and Master Forensic System

The lip-sync module uses a Lip-Sync Embedding Network (LSE-Net) inspired by SyncNet [16] and Wav2Lip [17]. Given paired audio and video streams, the module extracts mouth-region crops (via RetinaFace) and corresponding mel-spectrogram

windows, and embeds them into a shared latent space. Cosine similarity between the audio and visual vectors yields a lip-sync anomaly score. Figure 10 shows the master forensic system that integrates lip-sync with parallel visual experts (VideoMAE [35] and ViT) and audio experts (Wav2Vec2, CLAP, AST, onset analysis). A master decision logic compares fused scores against tuned thresholds and assigns the input to one of five forensic cases—Authentic, Sync Mismatch, Audio Fake, Video Fake, or Full Deepfake—and Claude 3.5 Sonnet produces the final forensic justification. Figure 11 shows a representative classification of Case 2 (Fake Video + Real Audio).

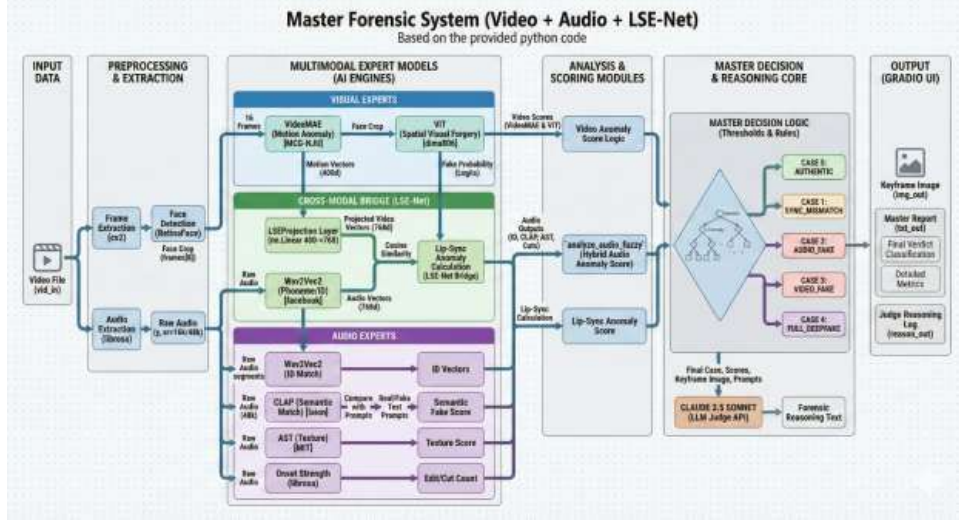


Fig. 10. Master forensic system combining visual experts (VideoMAE, ViT), audio experts (Wav2Vec2, CLAP, AST, onset analysis), and the LSE-Net cross-modal bridge. Outputs are fused through a master decision logic that distinguishes five forensic cases.

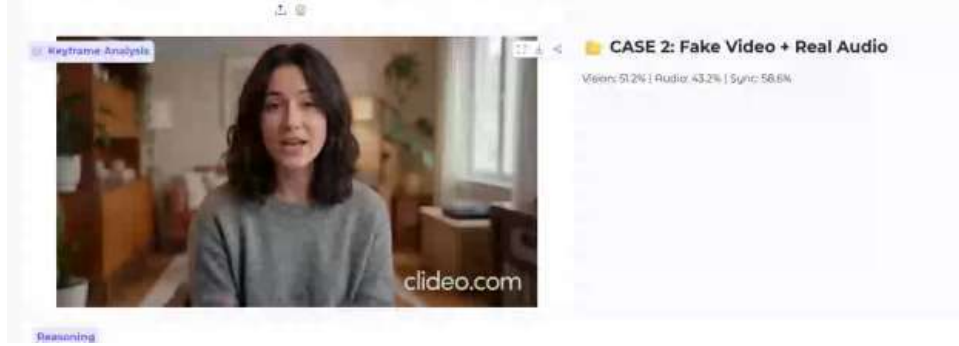


Fig. 11. Example output of the multimodal lip-sync module: the uploaded video is classified as Case 2 — Fake Video + Real Audio, with vision 51.2%, audio 43.2%, and sync 58.6%.

IV. DATASETS AND PREPROCESSING

Table I summarizes the datasets used in this study. FaceForensics++ [4] provides 1,000 video sequences with both pristine and four manipulation types (Deepfakes, Face2Face, FaceSwap, NeuralTextures); we use the c23 (high-quality) compression setting. FakeAVCeleb [10] supplies aligned audio-visual pairs of celebrity speakers; we sample a balanced subset with both real and synthesized lip-audio combinations. Our custom audio dataset combines public ASVspoof samples [5] with locally recorded utterances and synthetic samples generated by widely available text-to-speech systems. Text claims are retrieved at inference time from open-web sources. Standard preprocessing is applied: face detection and alignment via RetinaFace [33], voice activity detection for audio, sentence segmentation for text, and uniform 10-frame sampling for video. All trainable modules are optimized with AdamW ($\text{lr} = 2 \times 10^{-4}$, weight decay = 10^{-2}) with early stopping on validation F1.

TABLE I — Datasets and Modalities

Dataset	Modality	Source	Used For
FaceForensics++ [4]	Image / Video	Public benchmark (c23)	Image and video module training/evaluation

FakeAVCeleb [10]	Audio + Video (lip-sync)	Public benchmark	Lip-sync module training/evaluation
Custom Audio (incl. ASVspoof [5])	Audio (real + synthetic)	Public + custom recordings	Audio module training/evaluation
Web-retrieved claims (SERP)	Text	Real-time SERP API queries	Text module evidence at inference time

V. EXPERIMENTAL RESULTS

A. Per-Modality Performance

Table II reports test-set performance for each of the five modules. The text module achieves the highest accuracy (99%) due to the strong corroborative signal from LLM-based retrieval; lip-sync detection is the most challenging modality (89%), reflecting the difficulty of fine-grained cross-modal alignment under compression. Each modality is evaluated independently on its corresponding dataset, with binary classification metrics computed against held-out test splits.

TABLE II — Per-Modality Test Performance

Module	Accuracy	Precision	Recall	F1 Score
Text	99.0%	97.0%	98.0%	97.5%
Image	92.0%	90.0%	92.0%	91.0%
Audio	91.0%	89.0%	91.0%	90.0%
Video	91.0%	89.0%	91.0%	90.0%
Lip-Sync	89.0%	87.0%	89.0%	88.0%

B. Qualitative Outputs and Real-Time Deployment

Figures 3, 5, 7, 9, and 11 show representative qualitative outputs of each module on real inputs to the deployed system. The system is deployed publicly via Gradio on Hugging Face Spaces, providing reproducible access for external evaluation. Inference is performed on a single GPU and supports both file upload and live capture for each modality.

C. Note on Late-Fusion Evaluation

In the present work, each module is evaluated independently on its native modality. A unified late-fusion evaluation across coordinated multimodal attacks—where text, audio, and video are simultaneously manipulated within a single coherent forgery—requires a paired multimodal benchmark with coordinated ground-truth labels across all five streams, which to our knowledge is not yet publicly available. We therefore reserve quantitative late-fusion results for future work and focus the present evaluation on per-modality performance and qualitative deployment behavior.

VI. DISCUSSION

Three observations stand out from the experimental results. First, the modular per-modality design provides robust standalone performance across all five streams, with each module achieving accuracy above 89%, despite using publicly available pre-trained backbones rather than purpose-built architectures. Second, the text-verification module benefits substantially from retrieval-augmentation: grounding LLM stance classification in live web evidence provides a strong corroborative signal that pure LLM reasoning lacks. Third, lip-sync detection remains the bottleneck, indicating that fine-grained audio-visual alignment is still an open problem, especially when audio is generated by recent neural TTS systems trained on the same speakers. These observations align with recent findings on Deepfake-Eval-2024 [3] that academic deepfake datasets do not generalize well to in-the-wild manipulations. We therefore caution against over-interpreting the high per-modality accuracy figures reported here as indicative of real-world performance on unseen, coordinated multimodal attacks.

VII. LIMITATIONS AND THREATS TO VALIDITY

Several limitations of the current work merit explicit discussion.

- (i) The text-verification module depends on internet APIs (SERP, Groq) for evidence retrieval and LLM reasoning, which limits offline deployment and introduces variability based on retrieval quality. Performance therefore depends partly on factors outside the model itself (search-engine indexing, news availability, API rate limits).
- (ii) The system is computationally heavy and is not currently edge-deployable; full multimodal inference requires GPU acceleration. Quantitative latency benchmarking on multiple hardware configurations is left as future work.
- (iii) Per-modality results are reported on standard public benchmarks (FaceForensics++, FakeAVCeleb, ASVspoof). Performance on heavily compressed media or unseen manipulation types is expected to degrade based on prior reports in the literature [3]. We have not yet conducted systematic compression-robustness experiments on our system.
- (iv) The current evaluation does not include adversarial robustness testing. Recent work [36] shows that adversarially perturbed deepfakes can defeat detectors that perform well on clean inputs.
- (v) Demographic bias in the training data (FaceForensics++ skews toward Caucasian and East Asian faces) may produce uneven performance across populations [37]. We have not yet conducted disaggregated subgroup evaluation.
- (vi) A unified late-fusion evaluation on coordinated multimodal attacks (where text, audio, and video are simultaneously manipulated within a single coherent forgery) is not included in the present work due to the absence of a paired multimodal ground-truth benchmark. We outline this as a primary direction for future work.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a modular multimodal framework for fake-news and deepfake detection that integrates text, audio, image, video, and lip-sync analysis through five independent expert modules. The system achieves 89–99% per-modality accuracy on standard benchmarks and is deployed publicly on Hugging Face Spaces with explainable per-modality outputs. The principal design contribution is the modular architecture itself: each module can be retrained, replaced, or audited independently, and each verdict is accompanied by a natural-language explanation traceable to its underlying evidence. We argue that modularity and interpretability are essential—not optional—properties for real-world deployment by journalists, fact-checkers, and forensic analysts.

Future work will focus on four directions: (1) a unified late-fusion evaluation across paired multimodal attacks, requiring construction or curation of a coordinated-manipulation benchmark; (2) edge-deployable lightweight variants of each module to enable on-device detection in low-resource environments; (3) systematic adversarial-robustness evaluation through certified defenses and adversarial training; and (4) bias auditing of both the training datasets and the resulting detectors using established AI-accountability frameworks [38]. We also plan large-scale evaluation on Deepfake-Eval-2024 [3] and other in-the-wild benchmarks to better characterize real-world performance.

ACKNOWLEDGMENT

Special thanks to Sir Ziad Nayer for invaluable supervision and feedback throughout this work, and the Hugging Face team for hosting infrastructure, also thanks the curators of the FaceForensics++, FakeAVCeleb, and ASVspoof benchmarks, and my family—especially my husband, Ahmed Bilal Zafar—for their unwavering support.

Disclosure of AI-Assisted Writing: The author used a large language model (Claude, by Anthropic) for language polishing, structural editing, and formatting assistance during manuscript preparation. All technical content—including system design, architectural decisions, model selection, dataset curation, experimental setup, and analysis of results—is the author's original work. The author has read and verified the final manuscript and takes full responsibility for its content.

REFERENCES

- [1] I. Goodfellow, J. Pouget-Abadie, M. Mirza, et al., "Generative adversarial nets," in Proc. NeurIPS, 2014, pp. 2672–2680.
- [2] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in Proc. NeurIPS, vol. 33, 2020, pp. 6840–6851.
- [3] N. A. Chandra, R. Murtfeldt, L. Qiu, et al., "Deepfake-Eval-2024: A multi-modal in-the-wild benchmark of deepfakes circulated in 2024," arXiv preprint arXiv:2503.02857, 2025.
- [4] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," in Proc. IEEE/CVF International Conference on Computer Vision (ICCV), 2019, pp. 1–11.
- [5] M. Todisco, X. Wang, V. Vestman, M. Sahidullah, et al., "ASVspoof 2019: Future horizons in spoofed and fake audio detection," in Proc. INTERSPEECH, 2019, pp. 1008–1012.

- [6] J. Lv, Y. Gao, L. Li, et al., "Multi-modal fake news detection: A survey on deep learning techniques, advances, and problems," Springer Journal of Computational Social Systems, 2025.
- [7] M. C. Stamm, A. Croitoru, M. Marinescu, et al., "Deepfake media generation and detection in the generative AI era: A survey and outlook," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2024.
- [8] L. Verdoliva, "Media forensics and DeepFakes: An overview," IEEE Journal of Selected Topics in Signal Processing, vol. 14, no. 5, pp. 910–932, 2020.
- [9] S. Jia, R. Lyu, K. Zhao, Y. Chen, et al., "Can ChatGPT detect DeepFakes? A study of using multimodal large language models for media forensics," arXiv preprint arXiv:2403.14077, 2024.
- [10] H. Khalid, S. Tariq, M. Kim, and S. S. Woo, "FakeAVCeleb: A novel audio-video multimodal deepfake dataset," in Proc. NeurIPS Datasets and Benchmarks Track, 2021.
- [11] F. Pei, X. Zhang, et al., "Deepfake generation and detection: A benchmark and survey," arXiv preprint arXiv:2403.17881, 2024.
- [12] P. Liu, Q. Tao, and J. Zhou, "Evolving from single-modal to multi-modal facial deepfake detection: Progress and challenges," arXiv preprint arXiv:2406.06965, 2024.
- [13] S. M. Qureshi, A. Saeed, S. H. Almotiri, F. Ahmad, and M. A. Al Ghamdi, "Deepfake forensics: A survey of digital forensic methods for multimodal deepfake identification on social media," PeerJ Computer Science, vol. 10, e2037, 2024.
- [14] J. Frank, T. Eisenhofer, L. Schönherr, A. Fischer, D. Kolossa, and T. Holz, "Leveraging frequency analysis for deep fake image recognition," in Proc. ICML, 2020.
- [15] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in Proc. NeurIPS, 2020, pp. 12449–12460.
- [16] J. S. Chung and A. Zisserman, "Out of time: Automated lip sync in the wild (SyncNet)," in Proc. ACCV Workshops, 2016, pp. 251–263.
- [17] K. R. Prajwal, R. Mukhopadhyay, V. P. Namboodiri, and C. V. Jawahar, "A lip sync expert is all you need for speech to lip generation in the wild (Wav2Lip)," in Proc. ACM Multimedia, 2020.
- [18] A. Alsaeedi, A. AlMansour, and A. Jamal, "Audio-visual multimodal deepfake detection using emotional recognition," Int. J. Adv. Comput. Sci. Appl. (IJACSA), vol. 16, no. 6, 2025.
- [19] X. Zhang, J. Chu, J. Zhao, et al., "ERF-BA-TFD+: A multimodal model for audio-visual deepfake detection," arXiv preprint arXiv:2508.17282, 2025.
- [20] S. Li, Y. Wang, H. Chen, et al., "Multimodal learning for fake news detection in short videos," arXiv preprint arXiv:2509.15578, 2025.
- [21] M. Al-alshaqi, D. B. Rawat, and C. Liu, "A BERT-based multimodal framework for better fake news detection," Computers, MDPI, vol. 14, no. 6, 2025.
- [22] Y. Zhu, Y. Wang, and Z. Yu, "MFND: A large-scale multimodal fake news detection benchmark," in Proc. IJCAI, 2025.
- [23] Z. Cheng, H. Yang, Y. Xiong, and X. Hu, "Explainable AI for forensic speech authentication within cognitive and computational neuroscience," Frontiers in Neuroscience, vol. 19, 2025.
- [24] F.-A. Croitoru, A.-I. Hiji, V. Hondru, et al., "The tug-of-war between deepfake generation and detection," in Proc. Data-centric Machine Learning Research Workshop (DMLR) at ICML, 2024.
- [25] SerpAPI, "Google Search Results API documentation," [Online]. Available: <https://serpapi.com/>, 2024.
- [26] P. He, X. Liu, J. Gao, and W. Chen, "DeBERTa: Decoding-enhanced BERT with disentangled attention," in Proc. ICLR, 2021.
- [27] Meta AI, "Llama 3: Open foundation and instruction-tuned chat models," Meta AI Research, 2024.
- [28] A. Radford, J. W. Kim, T. Xu, et al., "Robust speech recognition via large-scale weak supervision (Whisper)," in Proc. ICML, 2023.
- [29] Y. Gong, Y.-A. Chung, and J. Glass, "AST: Audio Spectrogram Transformer," in Proc. INTERSPEECH, 2021, pp. 571–575.
- [30] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, "CLAP: Learning audio concepts from natural language supervision," in Proc. IEEE ICASSP, 2023.
- [31] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE CVPR, 2016, pp. 770–778.
- [32] A. Vaswani, N. Shazeer, N. Parmar, et al., "Attention is all you need," in Proc. NeurIPS, 2017, pp. 5998–6008.
- [33] J. Deng, J. Guo, E. Ververas, I. Kotsia, and S. Zafeiriou, "RetinaFace: Single-stage dense face localisation in the wild," in Proc. CVPR, 2020.
- [34] G. Ilharco, M. Wortsman, R. Wightman, et al., "OpenCLIP," Zenodo, 2021. doi:10.5281/zenodo.5143773.
- [35] Z. Tong, Y. Song, J. Wang, and L. Wang, "VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training," in Proc. NeurIPS, 2022.
- [36] B. Zhang, J. P. Zhou, I. Shumailov, and N. Papernot, "On attribution of deepfakes," arXiv preprint arXiv:2008.09194, 2020.
- [37] T. Reiss, B. Cavia, and Y. Hoshen, "Detecting deepfakes without seeing any," arXiv preprint arXiv:2311.01458, 2023.
- [38] S. Singh and A. Dhumane, "Unmasking digital deceptions: An integrative review of deepfake detection, multimedia forensics, and cybersecurity challenges," MethodsX, vol. 13, 103632, 2025.

- [39] A. Dosovitskiy, L. Beyer, A. Kolesnikov, et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in Proc. ICLR, 2021.
- [40] A. Radford, J. W. Kim, C. Hallacy, et al., "Learning transferable visual models from natural language supervision (CLIP)," in Proc. ICML, 2021, pp. 8748–8763.